Date:

	Outlook	Tempreture	Hummidity	Windy	Rain
1	Sunny	Hot	high	weak	No
2	Sunny	Mild	high	Strong	No
3	Claudy	Hot	high	weak	yes
4	Rainy	Mild	high	weak	yes
5	Rainy	Cool	Normal	weak	yes
6	Rainy	Cool	Normal	Strong	No
7	Claudy	Mild	Normal	Strong	yes
8	Sunny	Cool	Normal	weak	No

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entropy of
* Calculate the Prime attribute of Data set
Rain

$$\therefore -i \sum P_i \log_2 P_i$$

$$-\left(\frac{4}{8}\log_2\frac{4}{8} - \frac{4}{8}\log_2\frac{4}{8}\right)$$

$$R = -\frac{4}{8}\log_2\frac{4}{8} + \frac{4}{8}\log_2\frac{4}{8}$$

$$R = 0$$

Now calculate the entropy of Outlook =
for Sonny:-

$$= -\frac{0}{3}\log_2\left(\frac{0}{3}\right) + \frac{3}{3}\log_2\left(\frac{3}{3}\right)$$

$$= 0$$

for Cloudy:

$$-\frac{2}{2}\log_2\left(\frac{2}{2}\right) + \frac{0}{2}\log_2\left(\frac{0}{2}\right)$$

Cloudy = 0

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for Rainy :-

$$-\frac{2}{3}\log_2\left(\frac{2}{3}\right) + \frac{1}{3}\log_2\left(\frac{1}{3}\right)$$

Rainy = -0.138

Calculate the total entropy :-

$$\frac{3}{8}\log(0) = (0)$$

$$\frac{2}{8}\log(0) = (0)$$

$$\frac{3}{8}\log(-0.138) = -0.322$$

$$0 + 0 + (-0.322)$$

entropy = -0.322

Now Calculate the entropy of Tempreture

for hot :

$$-\frac{2}{2}\log_2\left(\frac{1}{2}\right) + \frac{1}{2}\log_2\left(\frac{1}{2}\right)$$

hot = 0

for Mild

$$-\frac{2}{3}\log_2\left(\frac{2}{3}\right) + \frac{1}{3}\log_2\left(\frac{1}{3}\right)$$

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Mild = -0.138

For Cool

$$-\frac{1}{3}\log_2\left(\frac{1}{3}\right) + \frac{2}{3}\log_2\left(\frac{2}{3}\right)$$

Cool = 0.138

Total entropy

$$\frac{2}{8}\log(0) = 0$$

$$\frac{3}{8}\log(0.138) = -0.322$$

$$\frac{3}{8}\log(0.138) = -0.322$$

entropy = 0 + (-0.322) + (-0.322)

entropy = -0.644

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